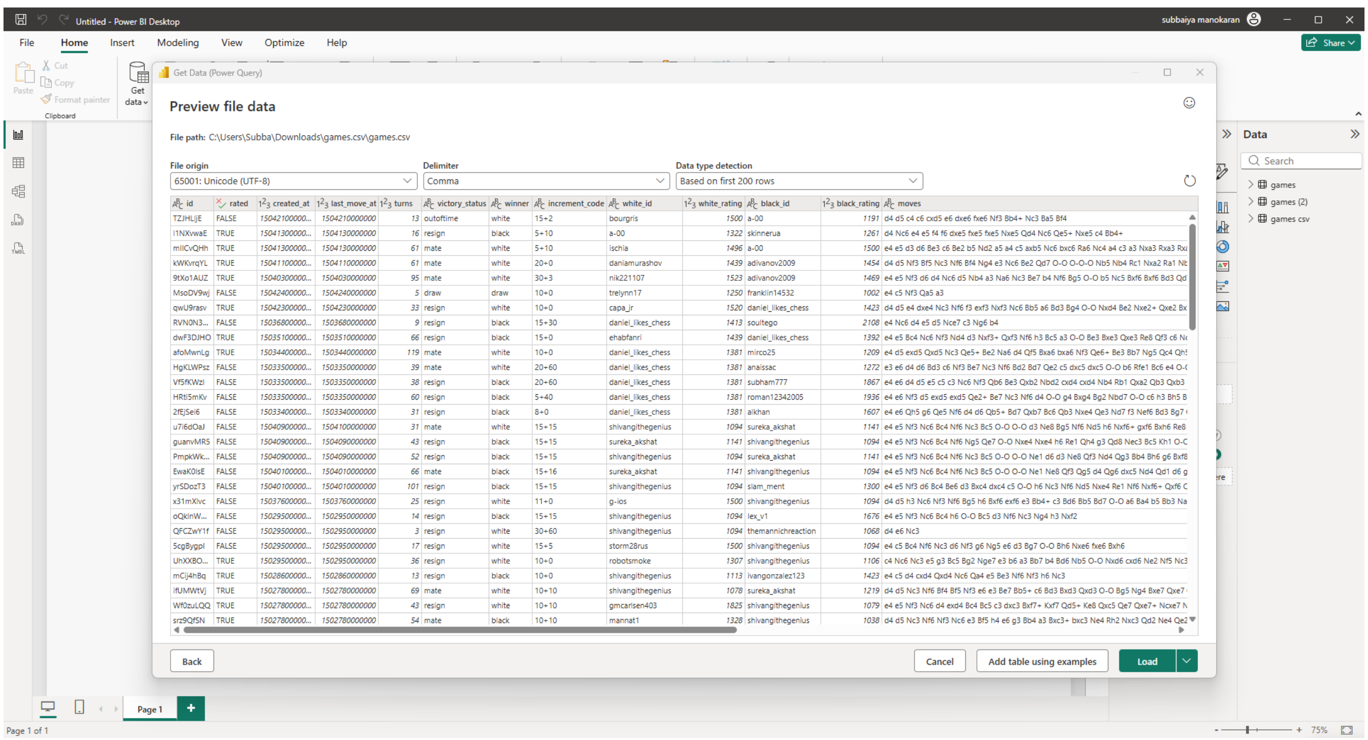
EXPERIMENT 6
DATA VISUALIZATION USING POWER BI

Aim

To import, transform, analyze, and visualize data using Power BI Desktop and create interactive dashboards and reports for decision-making.

Algorithm 1

Importing Data into Power BI
Step 1 Open Power BI Desktop
Step 2 Select: Home → Get Data
Step 3 Choose CSV
Step 4 Browse dataset and select it

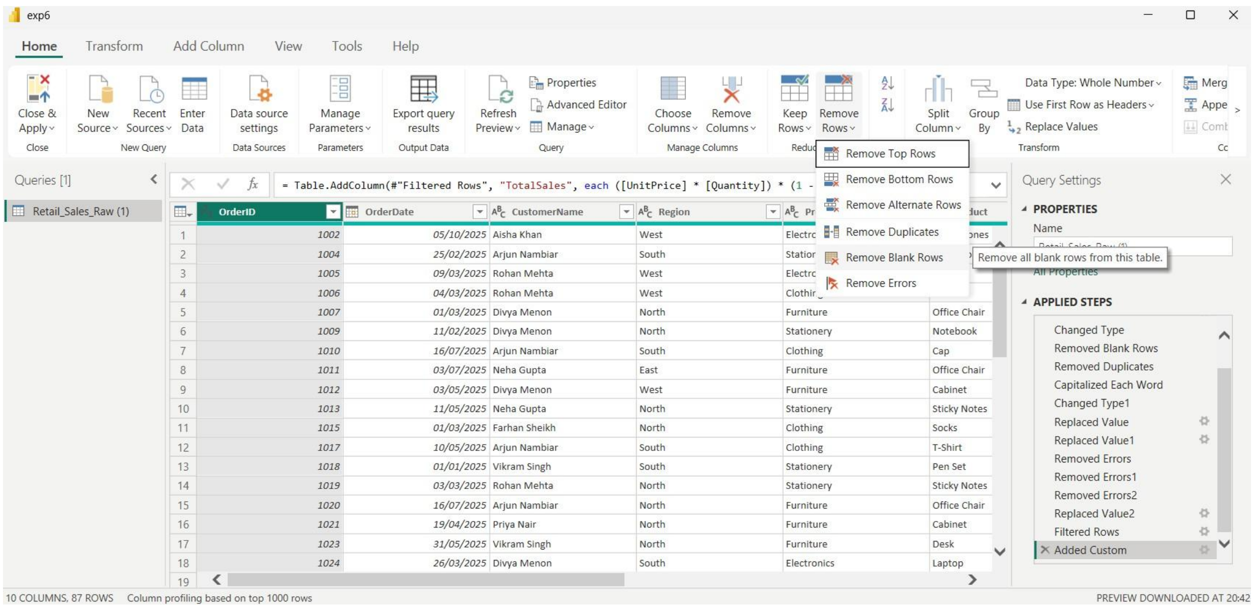


Algorithm 2

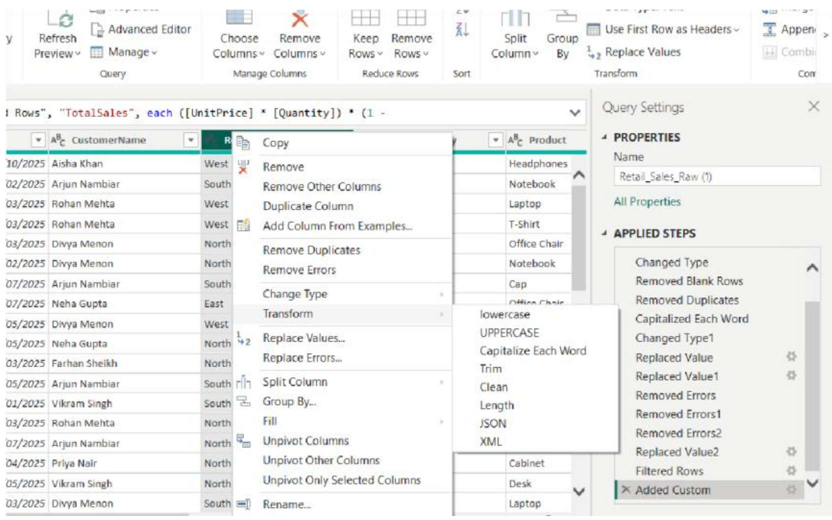
Data Transformation using Power Query
Step 1 Select Transform Data
Step 2 Open Power Query Editor
Step 3 Remove duplicates
Step 4 Handle null values
Step 5 Rename columns
Step 6 Apply changes

Procedure

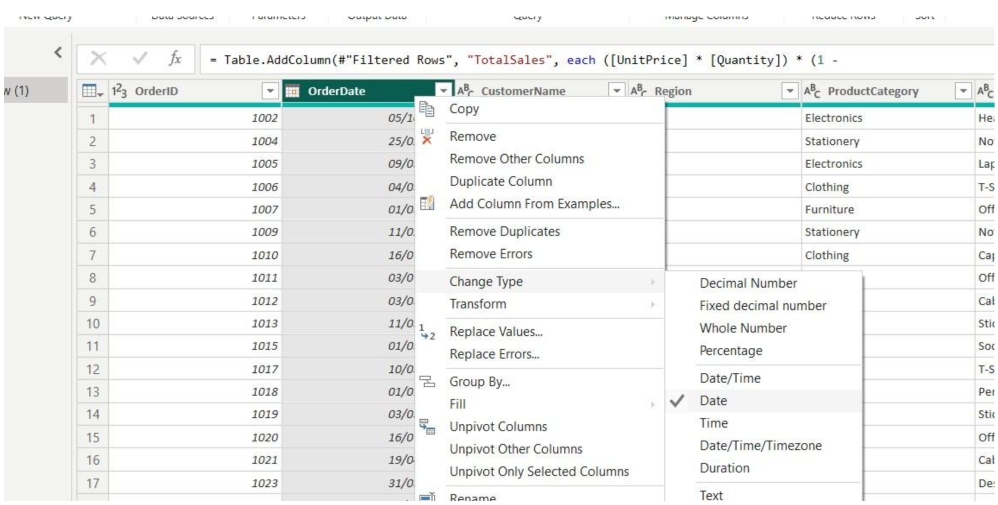
1. The data preview window will appear. Select Transform Data instead of Load to open the Power Query Editor.
2. In the Power Query Editor, examine the imported table and verify that all required columns are present, such as OrderID, OrderDate, CustomerName, Region, Quantity, UnitPrice, and Discount.
3. Remove blank rows from the dataset by selecting Home → Remove Rows → Remove Blank Rows. This removes completely empty records that could interfere with analysis and calculations.



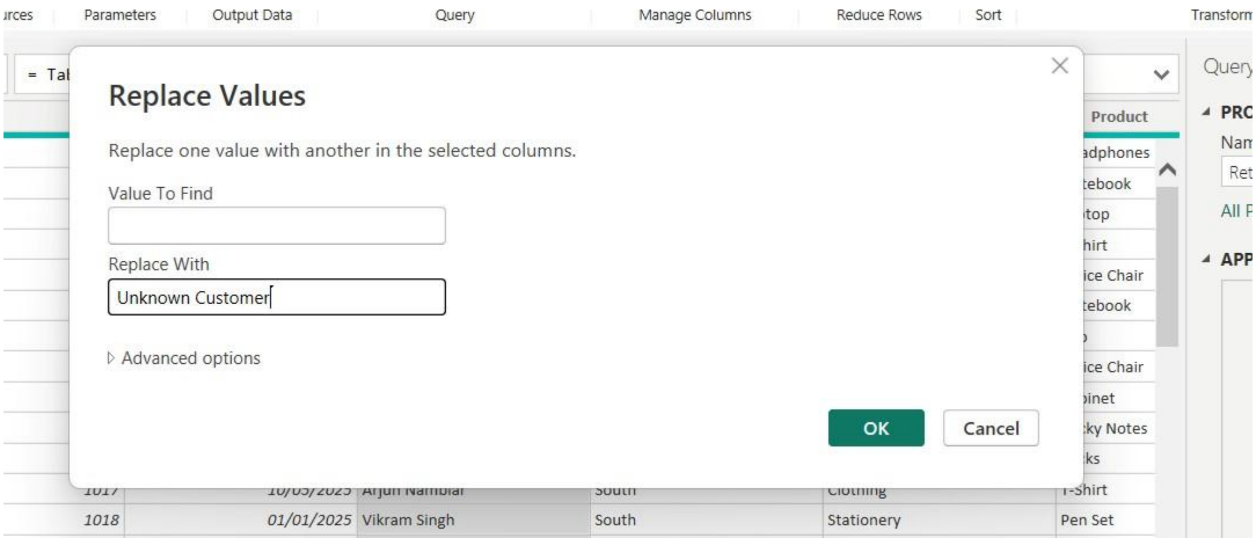
4. Remove duplicate records by selecting all columns in the table using Ctrl+A on the column headers and choosing Home → Remove Rows → Remove Duplicates. Power Query will remove records that contain identical values across all selected columns.
5. Standardize the values in the Region column. Right-click the Region column, select Transform, and choose Capitalize Each Word. This ensures that values such as “north”, “North”, and “NORTH” are represented consistently as “North”.



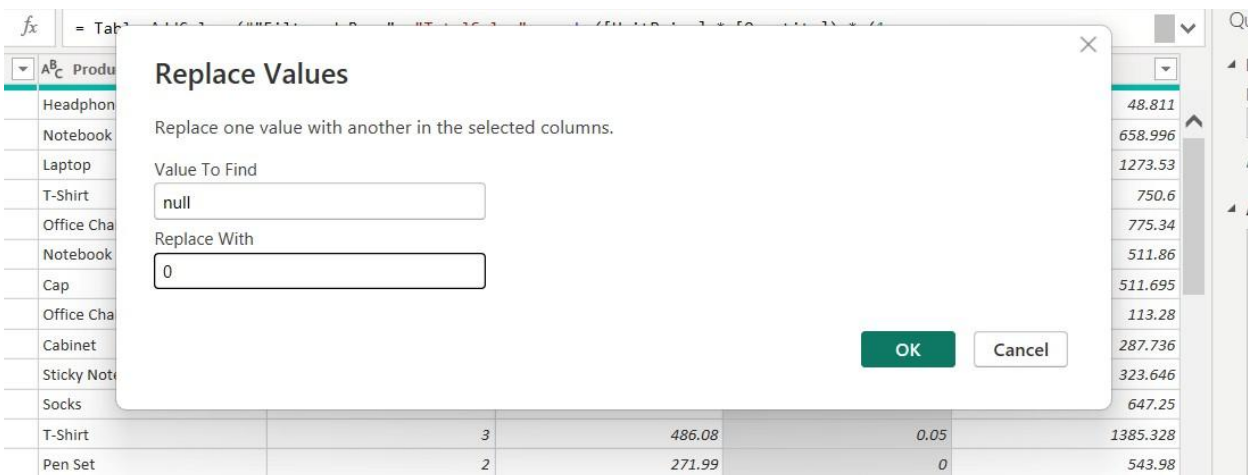
6. Standardize the OrderDate column by selecting it and choosing Transform → Data Type → Date. If Power Query displays errors because the dates use an inconsistent format, right-click the OrderDate column, select Change Type → Using Locale, and choose the appropriate date format and locale.



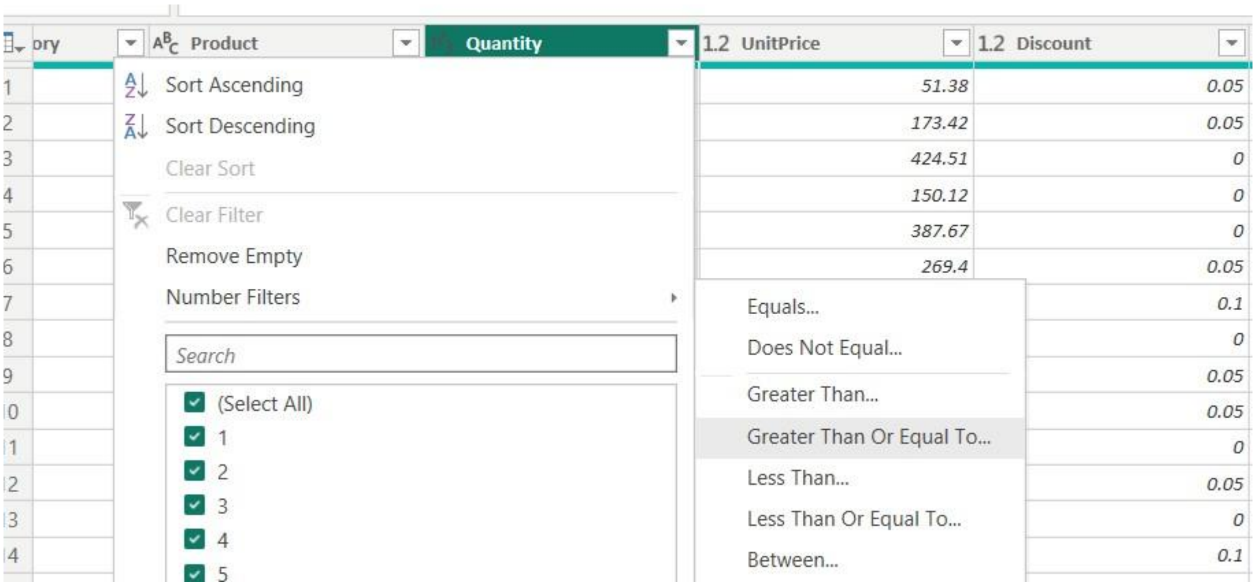
7. Handle missing values in the CustomerName column. Right-click CustomerName, select Replace Values, and replace blank or null values with “Unknown Customer”.

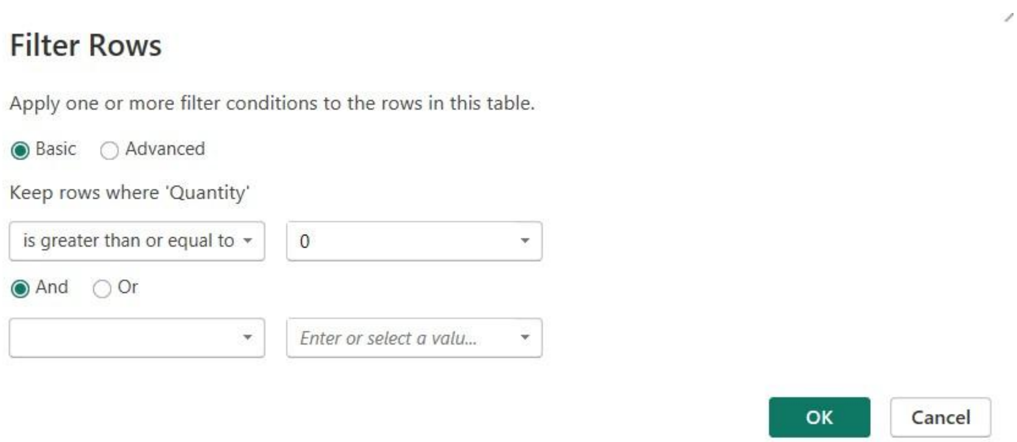


8. Handle missing values in the Discount column. Replace null values with 0 using the Replace Values option. A value of 0 represents that no discount was applied when the discount information is missing.

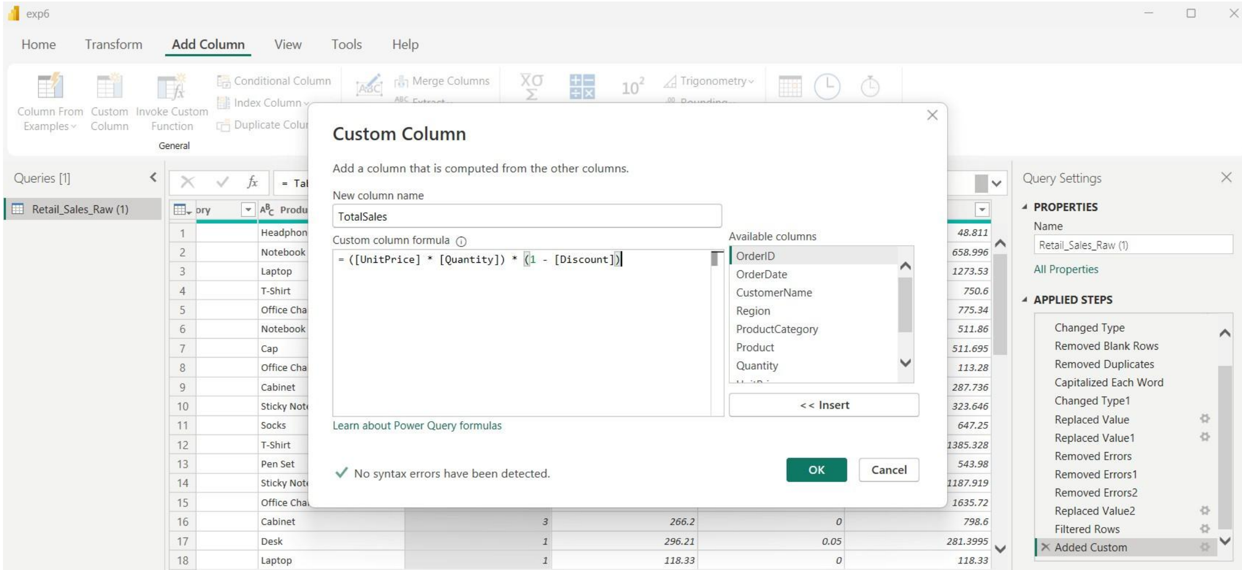


9. Identify and remove invalid Quantity values. Select the Quantity column and apply a filter by pressing the arrow next to the column name to exclude records where Quantity is less than 0. Negative quantities are treated as invalid business data because a normal sales transaction cannot contain a negative quantity.

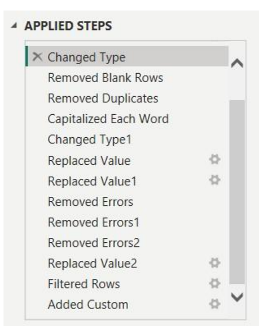




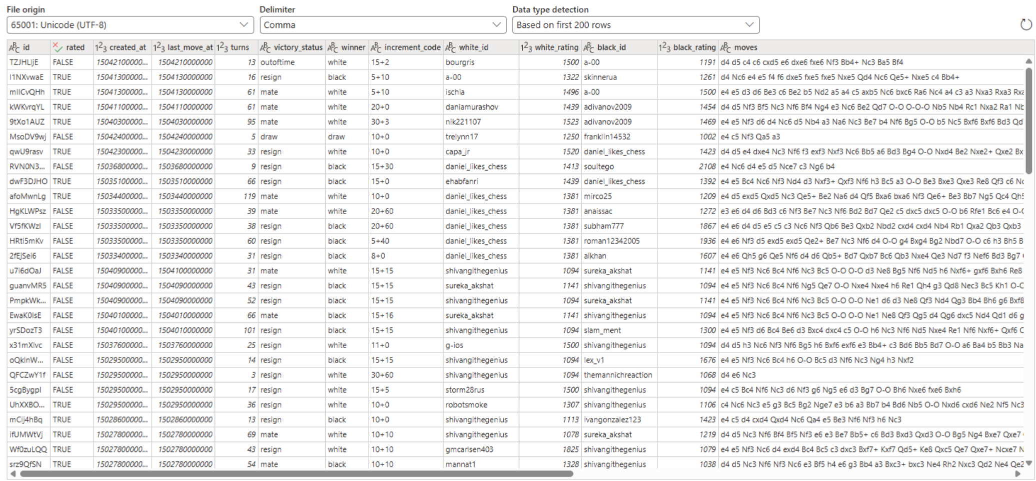
10. Create a derived column for total sales. Select Add Column → Custom Column and name the new column TotalSales. Enter the formula ([UnitPrice] * [Quantity]) * (1 - [Discount]). This calculates the actual sales value after applying the discount.



11. Verify and correct the data types of all columns. Set OrderID to Whole Number, OrderDate to Date, Quantity to Whole Number, UnitPrice to Decimal Number, Discount to Decimal Number, TotalSales to Decimal Number, and the remaining descriptive columns to Text.
12. Review the Applied Steps pane on the right side of the Power Query Editor. Confirm that all transformation steps have been recorded in the correct order. If an error is present, select the corresponding step and correct the transformation.



13. After completing all cleaning and transformation operations, select Home → Close & Apply. Power BI will apply the transformations and load the cleaned data into the data model.
14. Switch to the Data/Table view and inspect the transformed table. Verify that duplicate rows, blank rows, inconsistent Region values, invalid dates, missing customer names, missing discounts, and negative quantities have been handled correctly.



File origin: 65001: Unicode (UTF-8)
Delimiter: Comma
Data type detection: Based on first 200 rows

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l1NXvwaE	TRUE	15041300000...	1504130000000	16	resign	black	5+10	a-00	1322	skinnerua	1261	d4 Nc6 e4 e5 f4 f6 dxe5 fxe5 fxe5 Nxe5 Qd4 Nc6 Qe5+ Nxe5 c4 Bb4+
mIICvQHh	TRUE	15041300000...	1504130000000	61	mate	white	5+10	ischia	1496	a-00	1500	e4 e5 d3 d6 Be3 c6 Be2 b5 Nd2 a5 a4 c5 axb5 Nc6 bxc6 Ra6 Nc4 a4 c3 a3 Nxa3 Rxa3 Rxa
kWKvrqYL	TRUE	15041100000...	1504110000000	61	mate	white	20+0	daniamurashov	1439	adivanov2009	1454	d4 d5 Nf3 Bf5 Nc3 Nf6 Bf4 Ng4 e3 Nc6 Be2 Qd7 O-O O-O-O Nb5 Nb4 Rc1 Nxa2 Ra1 Nb
9tXo1AUZ	TRUE	15040300000...	1504030000000	95	mate	white	30+3	nik221107	1523	adivanov2009	1469	e4 e5 Nf3 d6 d4 Nc6 d5 Nb4 a3 Na6 Nc3 Be7 b4 Nf6 Bg5 O-O b5 Nc5 Bxf6 Bxf6 Bd3 Qd
MsoDV9wj	FALSE	15042400000...	1504240000000	5	draw	draw	10+0	trelynn17	1250	franklin14532	1002	e4 c5 Nf3 Qa5 a3
qwU9rasv	TRUE	15042300000...	1504230000000	33	resign	white	10+0	capa_jr	1520	daniel_likes_chess	1423	d4 d5 e4 dxe4 Nc3 Nf6 f3 exf3 Nxf3 Nc6 Bb5 a6 Bd3 Bg4 O-O Nxd4 Be2 Nxe2+ Qxe2 Bx
RVN0N3...	FALSE	15036800000...	1503680000000	9	resign	black	15+30	daniel_likes_chess	1413	soultego	2108	e4 Nc6 d4 e5 d5 Nce7 c3 Ng6 b4
dwF3DJHO	TRUE	15035100000...	1503510000000	66	resign	black	15+0	ehabfanri	1439	daniel_likes_chess	1392	e4 e5 Bc4 Nc6 Nf3 Nd4 d3 Nxf3+ Qxf3 Nf6 h3 Bc5 a3 O-O Be3 Bxe3 Qxe3 Re8 Qf3 c6 N
afoMwnLg	TRUE	15034400000...	1503440000000	119	mate	white	10+0	daniel_likes_chess	1381	mirco25	1209	e4 d5 exd5 Qxd5 Nc3 Qe5+ Be2 Na6 d4 Qf5 Bxa6 bxa6 Nf3 Qe6+ Be3 Bb7 Ng5 Qc4 Qh
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PmpkWk...	FALSE	15040900000...	1504090000000	52	resign	black	15+15	shivangithegenius	1094	sureka_akshat	1141	e4 e5 Nf3 Nc6 Bc4 Nf6 Nc3 Bc5 O-O O-O Ne1 d6 d3 Ne8 Qf3 Nd4 Qg3 Bb4 Bh6 g6 Bxf8
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x31mXlvc	FALSE	15037600000...	1503760000000	25	resign	white	11+0	g-ios	1500	shivangithegenius	1094	d4 d5 h3 Nc6 Nf3 Nf6 Bg5 h6 Bxf6 exf6 e3 Bb4+ c3 Bd6 Bb5 Bd7 O-O a6 Ba4 b5 Bb3 Na
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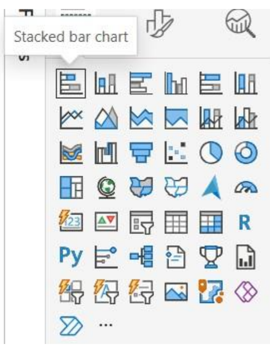
Algorithm 3

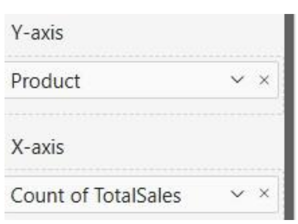
Creating a Bar Chart

Step 1 Choose Bar Chart Visualization

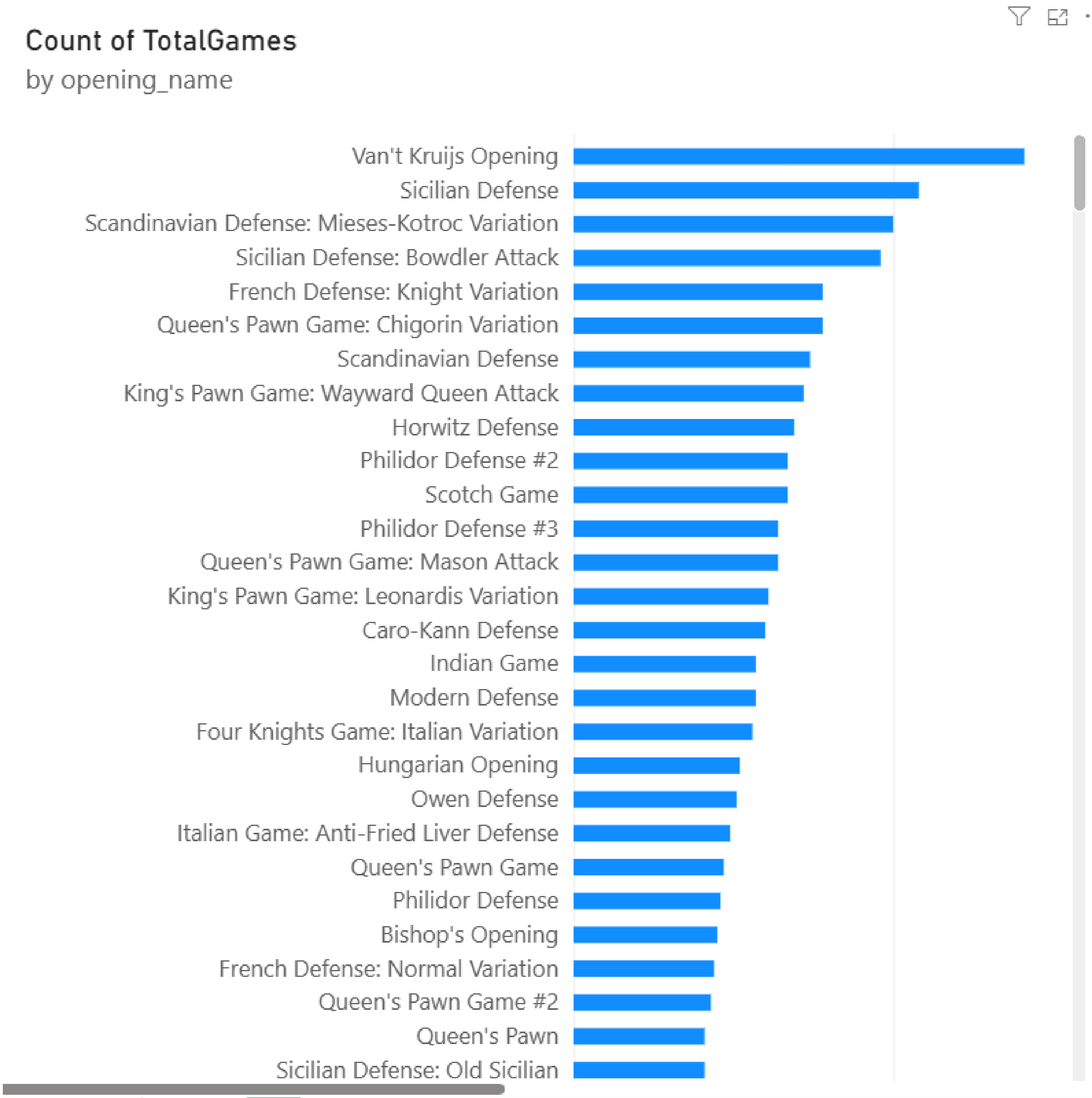


Step 2 Drag Product → Y-Axis

Step 3 Drag TotalSales → X axis



Step 4 Format chart



Interpretation:

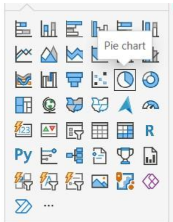
Cabinet generated highest sales

Stapler generated lowest sales

Algorithm 4

Creating a Pie Chart

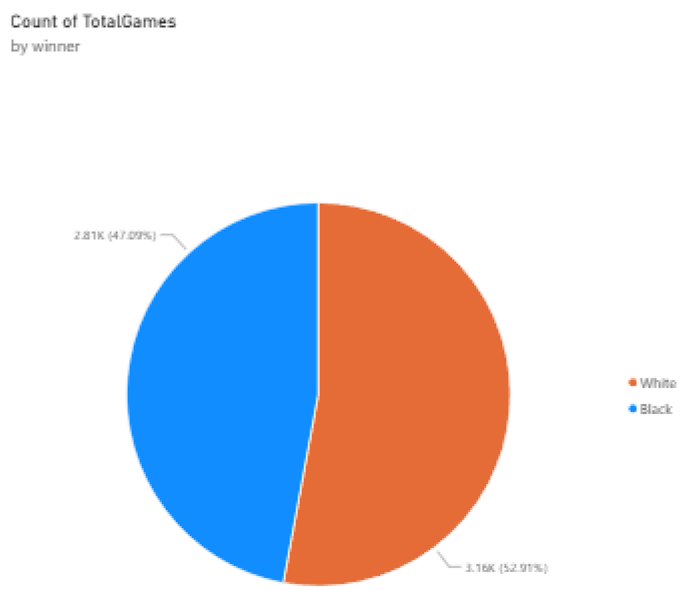
Step 1 Select Pie Chart



Step 2 Drag Product → Legend

Step 3 Drag TotalSales → Values

Output:



Interpretation:

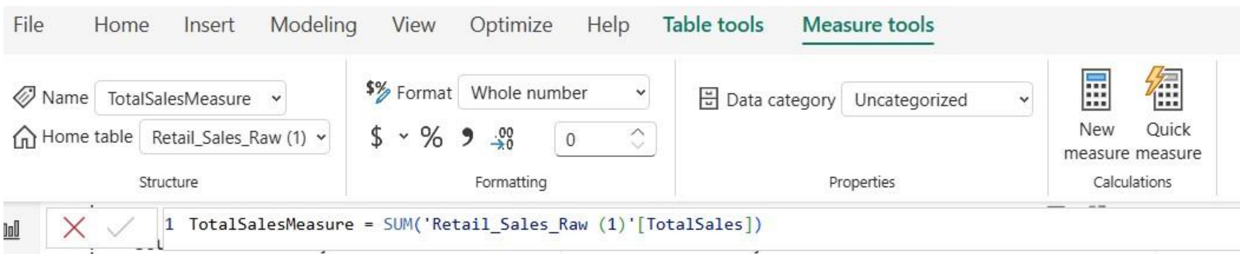
Cabinet generated highest Sales with 11.49%

Algorithm 5

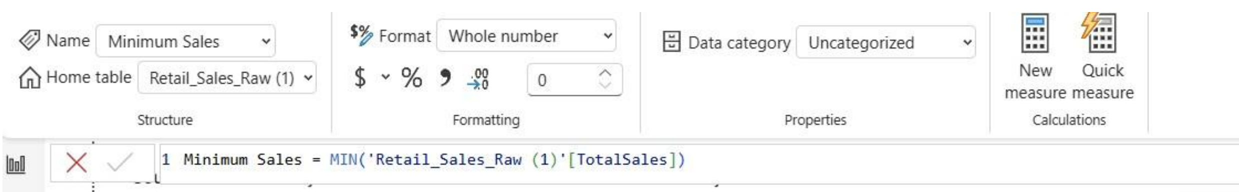
Creating a KPI Card

Step 1 Create 4 measures by clicking on the New Measure from Measure tools tab, type the formula inside the dax field:

a. TotalSalesMeasure

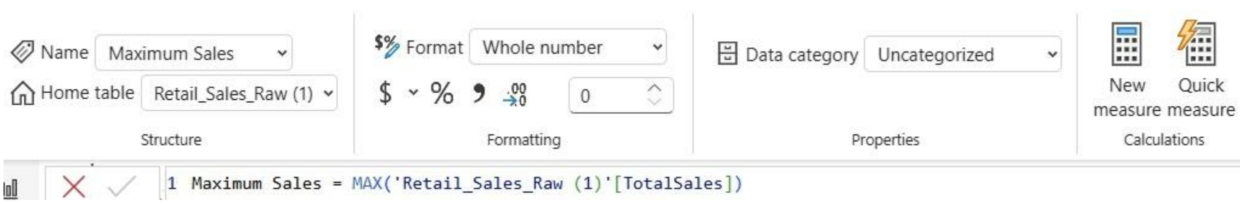


b. Minimum



c.

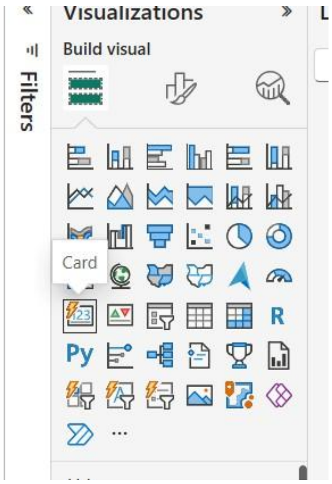
Maximum



d.

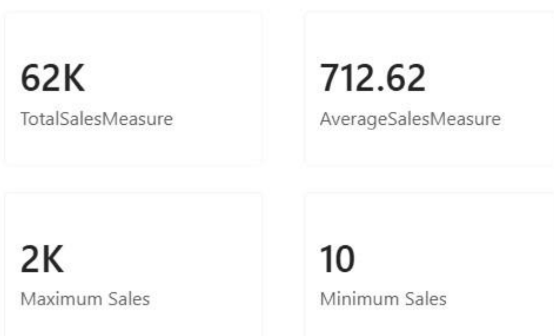
Average

Step 2 Create card visual



Step 3 Drag that measure into the field

Output:

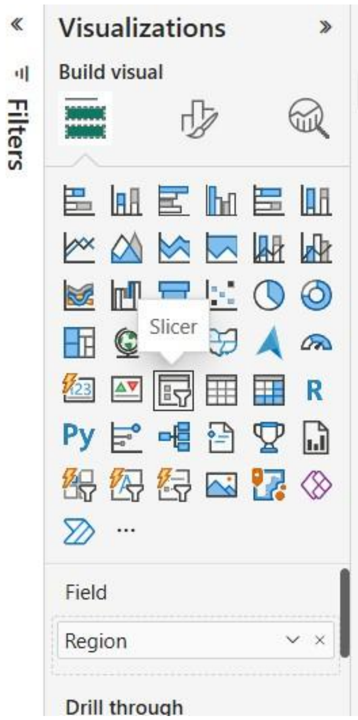


Algorithm 6

Creating Slicers

Step 1 Select Slicer Visualization

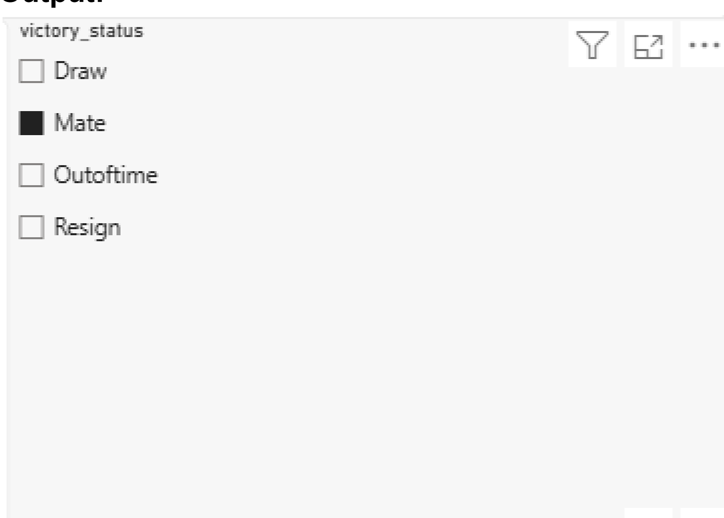
Step 2 Drag Region Field



Step 3 Enable Single Selection

Step 4 Apply Filters

Output:



Algorithm 8

Creating Dashboard

Step 1 Add KPI Card

Step 2 Add Bar Chart

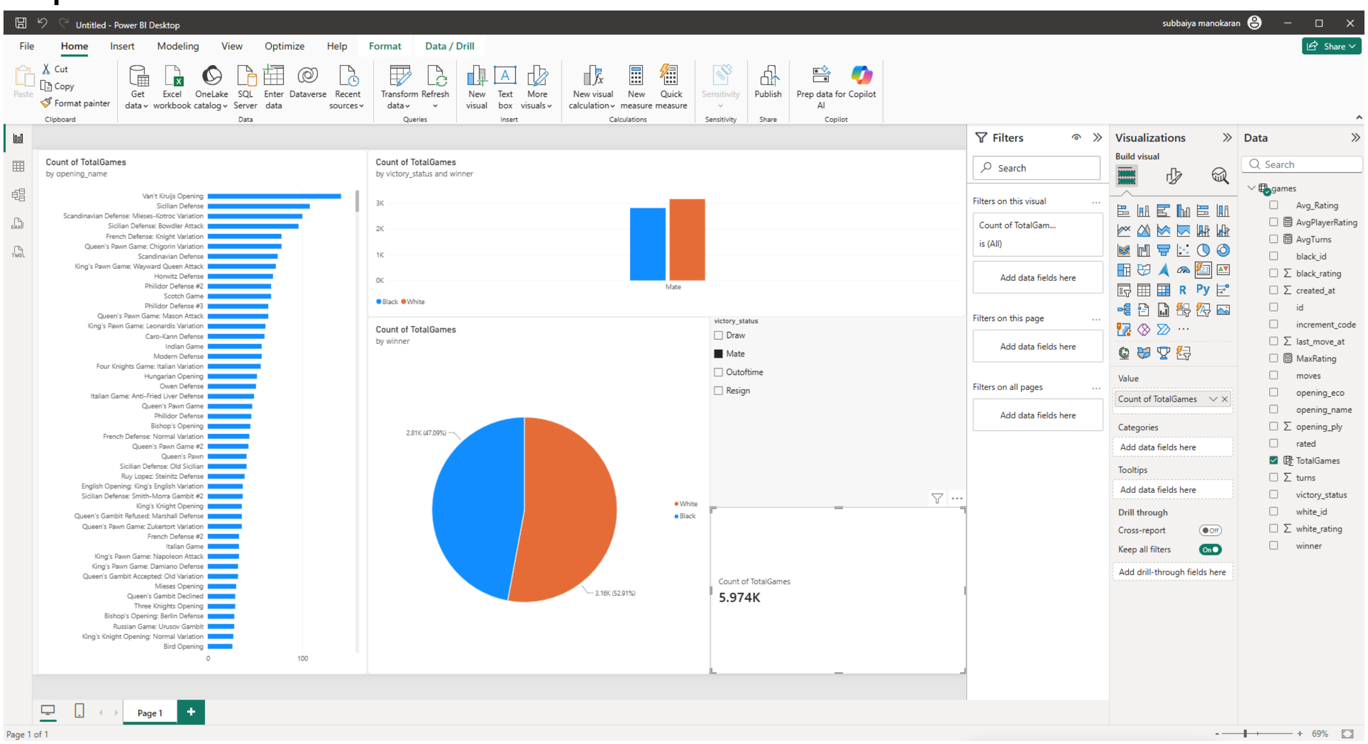
Step 3 Add Pie Chart

Step 4 Add Slicer

Step 5 Arrange Visuals

Step 6 Save Dashboard

Output:



Result:

The data was successfully imported, transformed, analyzed, and visualized in Power BI Desktop, and an interactive dashboard was created for effective decision-making.